KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4341 | 4 Credits: 4 | Program Core | Source-grounded

Mathematical Foundations for ML

 To provide a mathematical introduction in the field of Linear Algebra, calculus,

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4341 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4341.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Artificial Intelligence & Machine Learning
Course code	4341
Course title	Mathematical Foundations for ML
Semester	4 Credits: 4
Course category	Program Core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

20 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide a mathematical introduction in the field of Linear Algebra, calculus,
- probability and statistics to understand and apply algorithms in various applications.

Course outcomes

CO	Official outcome	Study level
CO1	implement Eigenvalues and Eigenvectors in 13 Applying engineering problems Utilize the concept related to partial derivatives to	See official syllabus
CO2	solve problems of maxima and minima of 15 Applying multivariable functions. Apply the concepts of probability for solving	See official syllabus
CO3	problems. 15 Applying Assess data using statistical methods and analyze	See official syllabus
CO4	that data by applying hypothesis testing methods. 15 Applying	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Eigenvectors in engineering problems	4	As specified
Module 2	maxima and minima of multivariable functions.	4	As specified
Module 3	Apply the concepts of probability for solving problems.	6	As specified
Module 4	hypothesis testing methods.	6	As specified

MODULE 1

Eigenvectors in engineering problems

This module is presented from the official syllabus scope for Mathematical Foundations for ML. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Eigenvectors in engineering problems
- Official duration: As specified

- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain the concept of vector spaces 2 Understanding Describe linearly independent and dependent

Explain the concept of vector spaces 2 Understanding Describe linearly independent and dependent is an official topic in Module 1 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the concept of vector spaces 2 Understanding Describe linearly independent and dependent using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the concept of vector spaces 2 understanding describe linearly independent and dependent should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the concept of vector spaces 2 Understanding Describe linearly independent and dependent means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain the concept of vector spaces 2 Understanding Describe linearly independent and dependent [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

2 Understanding vectors

2 Understanding vectors is an official topic in Module 1 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding vectors using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding vectors should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding vectors means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 2 Understanding vectors [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Compute the rank of a matrix 4 Applying Solve a system of linear equations by Gauss

Compute the rank of a matrix 4 Applying Solve a system of linear equations by Gauss is an official topic in Module 1 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compute the rank of a matrix 4 Applying Solve a system of linear equations by Gauss using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compute the rank of a matrix 4 applying solve a system of linear equations by gauss should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compute the rank of a matrix 4 Applying Solve a system of linear equations by Gauss means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Compute the rank of a matrix 4 Applying Solve a system of linear equations by Gauss □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

elimination method and evaluate the 5 Applying EigenValues and EigenVectors of a matrix Vector spaces: Vector spaces, linear independence, basis and rank, linear combination of vectors, linearly independent and dependent vectors, rank of a matrix Systems of linear equations:- Method of solution:- Gauss Elimination Method ,Eigenvalues and Eigenvectors and their properties. Utilize the concept related to partial derivatives to solve problems of

elimination method and evaluate the 5 Applying EigenValues and EigenVectors of a matrix Vector spaces: Vector spaces, linear independence, basis and rank, linear combination of vectors, linearly independent and dependent vectors, rank of a matrix Systems of linear equations:- Method of solution:- Gauss Elimination Method ,Eigenvalues and Eigenvectors and their properties. Utilize the concept related to partial derivatives to solve problems of is an official topic in Module 1 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify elimination method and evaluate the 5 Applying EigenValues and EigenVectors of a matrix Vector spaces: Vector spaces, linear independence, basis and rank, linear combination of vectors, linearly independent and dependent vectors, rank of a matrix Systems of linear equations:- Method of solution:- Gauss Elimination Method ,Eigenvalues and Eigenvectors and their properties. Utilize the concept related to partial derivatives to solve problems of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, elimination method and evaluate the 5 applying eigenvalues and eigenvectors of a matrix vector spaces: vector spaces, linear independence, basis and rank, linear combination of vectors, linearly independent and dependent vectors, rank of a matrix systems of linear equations:- method of solution:- gauss elimination method ,eigenvalues and eigenvectors and their properties. utilize the concept related to partial derivatives to solve problems of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What elimination method and evaluate the 5 Applying EigenValues and EigenVectors of a matrix Vector spaces: Vector spaces, linear independence, basis and rank, linear combination of vectors, linearly independent and dependent vectors, rank of a matrix Systems of linear equations:- Method of solution:- Gauss Elimination Method ,Eigenvalues and Eigenvectors and their properties. Utilize the concept related to partial derivatives to solve problems of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ elimination method and evaluate the 5 Applying EigenValues and EigenVectors of a matrix Vector spaces: Vector spaces, linear independence, basis and rank, linear combination of vectors, linearly independent and dependent vectors, rank of a matrix Systems of linear equations:- Method of solution:- Gauss Elimination Method ,Eigenvalues and Eigenvectors and their properties. Utilize the concept related to partial derivatives to solve problems of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Eigenvectors in engineering problems

   M1.01       Explain the concept of vector spaces                2
Understanding


               Describe linearly independent and dependent
   M1.02                                                           2
Understanding
               vectors

   M1.03       Compute the rank of a matrix                        4
Applying


               Solve a system of linear equations by Gauss
   M1.04       elimination method and evaluate the                 5
Applying
               EigenValues and EigenVectors of a matrix
 Contents:
 Vector spaces: Vector spaces, linear independence, basis and rank, linear
combination of
 vectors, linearly independent and dependent vectors, rank of a matrix
 Systems of linear equations:- Method of solution:- Gauss Elimination Method
,Eigenvalues
 and Eigenvectors and their properties.

               Utilize the concept related to partial derivatives to solve
problems of
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

maxima and minima of multivariable functions.

This module is presented from the official syllabus scope for Mathematical Foundations for ML. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: maxima and minima of multivariable functions.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Applying of functions of two variables Outline the Concept of first and second order

4 Applying of functions of two variables Outline the Concept of first and second order is an official topic in Module 2 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying of functions of two variables Outline the Concept of first and second order using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying of functions of two variables outline the concept of first and second order should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying of functions of two variables Outline the Concept of first and second order means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 4 Applying of functions of two variables Outline the Concept of first and second order [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Applying Partial derivatives Explain the Concept of maxima and minima of

4 Applying Partial derivatives Explain the Concept of maxima and minima of is an official topic in Module 2 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying Partial derivatives Explain the Concept of maxima and minima of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying partial derivatives explain the concept of maxima and minima of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying Partial derivatives Explain the Concept of maxima and minima of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-[illegible] 4 Applying Partial derivatives Explain the Concept of maxima and minima of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

functions of two variables and apply them in 4 Applying related problems

functions of two variables and apply them in 4 Applying related problems is an official topic in Module 2 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify functions of two variables and apply them in 4 Applying related problems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, functions of two variables and apply them in 4 applying related problems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What functions of two variables and apply them in 4 Applying related problems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-□□ functions of two variables and apply them in 4 Applying related problems □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Demonstrate Multivariate differentiation 3 Applying Calculus: Introduction to Functions, Limits and Continuity, Differentiability, Derivative rules and operations, partial derivatives, maxima and minima of functions of two variables, Multivariate differentiation (Simple problems based on these topics).

Demonstrate Multivariate differentiation 3 Applying Calculus: Introduction to Functions, Limits and Continuity, Differentiability, Derivative rules and operations, partial derivatives, maxima and minima of functions of two variables, Multivariate differentiation (Simple problems based on these topics). is an official topic in Module 2 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate Multivariate differentiation 3 Applying Calculus: Introduction to Functions, Limits and Continuity, Differentiability, Derivative rules and operations, partial derivatives, maxima and minima of functions of two variables, Multivariate differentiation (Simple problems based on these topics). using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate multivariate differentiation 3 applying calculus: introduction to functions, limits and continuity, differentiability, derivative rules and operations, partial derivatives, maxima and minima of functions of two variables, multivariate differentiation (simple problems based on these topics). should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate Multivariate differentiation 3 Applying Calculus: Introduction to Functions, Limits and Continuity, Differentiability, Derivative rules and operations, partial derivatives, maxima and minima of functions of two variables, Multivariate differentiation (Simple problems based on these topics). means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Demonstrate Multivariate differentiation 3 Applying Calculus: Introduction to Functions, Limits and Continuity, Differentiability, Derivative rules and operations, partial derivatives, maxima and minima of functions of two variables, Multivariate differentiation (Simple problems based on these topics). □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
maxima and minima of multivariable functions.

               Illustrate the Concept of limit and continuity
   M2.01                                                          4
Applying
               of functions of two variables

               Outline the Concept of first and second order
   M2.02                                                          4
Applying
               Partial derivatives

               Explain the Concept of maxima and minima of
   M2.03       functions of two variables and apply them in       4
Applying
               related problems
  M2.04       Demonstrate Multivariate differentiation            3
Applying

              Series Test – I                                     1

Contents:
Calculus: Introduction to Functions, Limits and Continuity, Differentiability,
Derivative
rules and operations, partial derivatives, maxima and minima of functions of two
variables,
Multivariate differentiation (Simple problems based on these topics).
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Apply the concepts of probability for solving problems.

This module is presented from the official syllabus scope for Mathematical Foundations for ML. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Apply the concepts of probability for solving problems.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

2 Applying space, dependent and independent events Illustrate conditional probability and Bayes’

2 Applying space, dependent and independent events Illustrate conditional probability and Bayes’ is an official topic in Module 3 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying space, dependent and independent events Illustrate conditional probability and Bayes’ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying space, dependent and independent events illustrate conditional probability and bayes’ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying space, dependent and independent events Illustrate conditional probability and Bayes’ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Applying space, dependent and independent events Illustrate conditional probability and Bayes’ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Applying Theorem Compare the difference between Discrete and

3 Applying Theorem Compare the difference between Discrete and is an official topic in Module 3 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying Theorem Compare the difference between Discrete and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying theorem compare the difference between discrete and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying Theorem Compare the difference between Discrete and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ 3 Applying Theorem Compare the difference between Discrete and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

3 Applying continuous random variables Calculate the expected value and variance of

3 Applying continuous random variables Calculate the expected value and variance of is an official topic in Module 3 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying continuous random variables Calculate the expected value and variance of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying continuous random variables calculate the expected value and variance of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying continuous random variables Calculate the expected value and variance of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Applying continuous random variables Calculate the expected value and variance of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Applying discrete and continuous random variables Identify the characteristics of discrete

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Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying discrete and continuous random variables Identify the characteristics of discrete using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying discrete and continuous random variables identify the characteristics of discrete should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying discrete and continuous random variables Identify the characteristics of discrete means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 3-□□ 3 Applying discrete and continuous random variables Identify the characteristics of discrete □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

probability distributions- binomial and 2 Understanding poisson Use probability distributions to solve simple

probability distributions- binomial and 2 Understanding poisson Use probability distributions to solve simple is an official topic in Module 3 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify probability distributions- binomial and 2 Understanding poisson Use probability distributions to solve simple using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, probability distributions- binomial and 2 understanding poisson use probability distributions to solve simple should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What probability distributions- binomial and 2 Understanding poisson Use probability distributions to solve simple means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] probability distributions- binomial and 2 Understanding poisson Use probability distributions to solve simple [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

2 Applying problems Probability: Axiomatic definition - sample space, dependent and independent events conditional probability,Bayes’ Theorem Random variables: Discrete and continuous random variables, expectation, variance, Probability distributions- binomial, poisson Assess data using statistical methods and analyze that data by applying

2 Applying problems Probability: Axiomatic definition - sample space, dependent and independent events conditional probability,Bayes’ Theorem Random variables: Discrete and continuous random variables, expectation, variance, Probability distributions- binomial, poisson Assess data using statistical methods and analyze that data by applying is an official topic in Module 3 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying problems Probability: Axiomatic definition - sample space, dependent and independent events conditional probability,Bayes’ Theorem Random variables: Discrete and continuous random variables, expectation, variance, Probability distributions- binomial, poisson Assess data using statistical methods and analyze that data by applying using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying problems probability: axiomatic definition - sample space, dependent and independent events conditional probability,bayes’ theorem random variables: discrete and continuous random variables, expectation, variance, probability distributions- binomial, poisson assess data using statistical methods and analyze that data by applying should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying problems Probability: Axiomatic definition - sample space, dependent and independent events conditional probability,Bayes’ Theorem Random variables: Discrete and continuous random variables, expectation, variance, Probability distributions- binomial, poisson Assess data using statistical methods and analyze that data by applying means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 3-□□ 2 Applying problems Probability: Axiomatic definition - sample space, dependent and independent events conditional probability,Bayes’ Theorem Random variables: Discrete and continuous random variables, expectation, variance, Probability distributions- binomial, poisson Assess data using statistical methods and analyze that data by applying □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Apply the concepts of probability for solving problems.

              Demonstrate the basic concepts in sample
  M3.01                                                          2
Applying
              space, dependent and independent events
              Illustrate conditional probability and Bayes'
  M3.02                                                          3
Applying
              Theorem
              Compare the difference between Discrete and
  M3.03                                                          3
Applying
              continuous random variables
              Calculate the expected value and variance of
  M3.04                                                          3
Applying
              discrete and continuous random variables
              Identify the characteristics of discrete
  M3.05       probability distributions- binomial and            2
Understanding
              poisson
              Use probability distributions to solve simple
  M3.06                                                          2
Applying
              problems
Contents:
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

hypothesis testing methods.

This module is presented from the official syllabus scope for Mathematical Foundations for ML. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: hypothesis testing methods.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

1 Understanding performed on data Calculate the measures of central tendency and

1 Understanding performed on data Calculate the measures of central tendency and is an official topic in Module 4 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding performed on data Calculate the measures of central tendency and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding performed on data calculate the measures of central tendency and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding performed on data Calculate the measures of central tendency and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 1 Understanding performed on data Calculate the measures of central tendency and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Applying dispersion Explain the concept of Sampling and Sampling

3 Applying dispersion Explain the concept of Sampling and Sampling is an official topic in Module 4 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying dispersion Explain the concept of Sampling and Sampling using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying dispersion explain the concept of sampling and sampling should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying dispersion Explain the concept of Sampling and Sampling means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 4-□□ 3 Applying dispersion Explain the concept of Sampling and Sampling □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

3 Understanding Distributions

3 Understanding Distributions is an official topic in Module 4 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Distributions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding distributions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Distributions means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 3 Understanding Distributions [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe the types of errors

Describe the types of errors is an official topic in Module 4 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the types of errors using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the types of errors should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the types of errors means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Describe the types of errors [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Calculate confidence interval for sample mean

Calculate confidence interval for sample mean is an official topic in Module 4 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Calculate confidence interval for sample mean using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, calculate confidence interval for sample mean should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Calculate confidence interval for sample mean means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Calculate confidence interval for sample mean [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Demonstrate the concept of hypothesis testing 3 Applying Fundamentals of Data: Collection, Summarization, and Visualization, Measures of central tendency and dispersion,Sampling and Sampling Distributions, Types of Errors, level of significance, Confidence Interval Estimation and Hypothesis Testing for Z -test and t- Test. Text / Reference: T/R Book Title / Author T1 H. Anton, I. Biven,S.Davis, “Calculus”, Wiley, 10th edition, 2015. T2 Mathematics for Machine Learning by Marc Peter Deisenroth,A.Aldo Faisal,and Cheng Soon Ong published by CAmbridge university press T3 Linear Algebra and its applications,4th edition by Gilbert Strang T4 Introduction to Applied Linear Algebra by Stephen Boyd and Lieven Vandenberghe,2018 published by Cambridge university Press Jay L.Devore,probability and Statistics form Engineering and the ScienceS,8th T5 edition,Cengage,2012 Online Resources: Sl. No. Website Link 1 https://www.w3schools.com/ai/ai_mathematics.asp 2 https://mml-book.github.io/book/mml-book.pdf

Demonstrate the concept of hypothesis testing 3 Applying Fundamentals of Data: Collection, Summarization, and Visualization, Measures of central tendency and dispersion,Sampling and Sampling Distributions, Types of Errors, level of significance, Confidence Interval Estimation and Hypothesis Testing for Z -test and t- Test. Text / Reference: T/R Book Title / Author T1 H. Anton, I. Biven,S.Davis, “Calculus”, Wiley, 10th edition, 2015. T2 Mathematics for Machine Learning by Marc Peter Deisenroth,A.Aldo Faisal,and Cheng Soon Ong published by CAmbridge university press T3 Linear Algebra and its applications,4th edition by Gilbert Strang T4 Introduction to Applied Linear Algebra by Stephen Boyd and Lieven Vandenberghe,2018 published by Cambridge university Press Jay L.Devore,probability and Statistics form Engineering and the ScienceS,8th T5 edition,Cengage,2012 Online Resources: Sl. No. Website Link 1 https://www.w3schools.com/ai/ai_mathematics.asp 2 https://mml-book.github.io/book/mml-book.pdf is an official topic in Module 4 of Mathematical Foundations for ML. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate the concept of hypothesis testing 3 Applying Fundamentals of Data: Collection, Summarization, and Visualization, Measures of central tendency and dispersion,Sampling and Sampling Distributions, Types of Errors, level of significance, Confidence Interval Estimation and Hypothesis Testing for Z -test and t- Test. Text / Reference: T/R Book Title / Author T1 H. Anton, I. Biven,S.Davis, “Calculus”, Wiley, 10th edition, 2015. T2 Mathematics for Machine Learning by Marc Peter Deisenroth,A.Aldo Faisal,and Cheng Soon Ong published by CAmbridge university press T3 Linear Algebra and its applications,4th edition by Gilbert Strang T4 Introduction to Applied Linear Algebra by Stephen Boyd and Lieven Vandenberghe,2018 published by Cambridge university Press Jay L.Devore,probability and Statistics form Engineering and the ScienceS,8th T5 edition,Cengage,2012 Online Resources: Sl. No. Website Link 1 https://www.w3schools.com/ai/ai_mathematics.asp 2 https://mml-book.github.io/book/mml-book.pdf using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate the concept of hypothesis testing 3 applying fundamentals of data: collection, summarization, and visualization, measures of central tendency and dispersion,sampling and sampling distributions, types of errors, level of significance, confidence interval estimation and hypothesis testing for z -test and t-test. text / reference: t/r book title / author t1 h. anton, i. biven,s.davis, “calculus”, wiley, 10th edition, 2015. t2 mathematics for machine learning by marc peter deisenroth,a.aldo faisal,and cheng soon ong published by cambridge university press t3 linear algebra and its applications,4th edition by gilbert strang t4 introduction to applied linear algebra by stephen boyd and lieven vandenberghe,2018 published by cambridge university press jay l.devore,probability and statistics form engineering and the sciences,8th t5 edition,cengage,2012 online resources: sl. no. website link 1 https://www.w3schools.com/ai/ai_mathematics.asp 2 https://mml-book.github.io/book/mml-book.pdf should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate the concept of hypothesis testing 3 Applying Fundamentals of Data: Collection, Summarization, and Visualization, Measures of central tendency and dispersion,Sampling and Sampling Distributions, Types of Errors, level of significance, Confidence Interval Estimation and Hypothesis Testing for Z -test and t- Test. Text / Reference: T/R Book Title / Author T1 H. Anton, I. Biven,S.Davis, “Calculus”, Wiley, 10th edition, 2015. T2 Mathematics for Machine Learning by Marc Peter Deisenroth,A.Aldo Faisal,and Cheng Soon Ong published by CAmbridge university press

Aspect	What to prepare
	T3 Linear Algebra and its applications,4th edition by Gilbert Strang T4 Introduction to Applied Linear Algebra by Stephen Boyd and Lieven Vandenberghe,2018 published by Cambridge university Press Jay L.Devore,probability and Statistics form Engineering and the ScienceS,8th T5 edition,Cengage,2012 Online Resources: Sl. No. Website Link 1 https://www.w3schools.com/ai/ai_mathematics.asp 2 https://mml-book.github.io/book/mml-book.pdf means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 4-□□ Demonstrate the concept of hypothesis testing 3 Applying Fundamentals of Data: Collection, Summarization, and Visualization, Measures of central tendency and dispersion,Sampling and Sampling Distributions, Types of Errors, level of significance, Confidence Interval Estimation and Hypothesis Testing for Z -test and t- Test. Text / Reference: T/R Book Title / Author T1 H. Anton, I. Biven,S.Davis, “Calculus”, Wiley, 10th edition, 2015. T2 Mathematics for Machine Learning by Marc Peter Deisenroth,A.Aldo Faisal,and Cheng Soon Ong published by CAmbridge university press T3 Linear Algebra and its applications,4th edition by Gilbert Strang T4 Introduction to Applied Linear Algebra by Stephen Boyd and Lieven Vandenberghe,2018 published by Cambridge university Press Jay L.Devore,probability and Statistics form Engineering and the ScienceS,8th T5 edition,Cengage,2012 Online Resources: Sl. No. Website Link 1 https://www.w3schools.com/ai/ai_mathematics.asp 2 https://mml-book.github.io/book/mml-book.pdf □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
hypothesis testing methods.

              Describe the fundamental operations to be
  M4.01                                                            1
Understanding
              performed on data
              Calculate the measures of central tendency and
  M4.02                                                            3
Applying
              dispersion
              Explain the concept of Sampling and Sampling
  M4.03                                                            3
Understanding
              Distributions
  M4.04       Describe the types of errors                         2
Understanding

  M4.05       Calculate confidence interval for sample mean        3
Applying

  M4.06       Demonstrate the concept of hypothesis testing        3
Applying
                 Series Test – II                                    1
 Contents:
 Fundamentals of Data: Collection, Summarization, and Visualization, Measures of
central
 tendency and dispersion,Sampling and Sampling Distributions, Types of Errors,
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Explain the concept of vector spaces 2 Understanding Describe linearly independent and dependent. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the concept of vector spaces 2 Understanding Describe linearly independent and dependent*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding vectors. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding vectors*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Compute the rank of a matrix 4 Applying Solve a system of linear equations by Gauss. (3 marks)

Answer: Begin with the standard definition or identity of *Compute the rank of a matrix 4 Applying Solve a system of linear equations by Gauss*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain elimination method and evaluate the 5 Applying EigenValues and EigenVectors of a matrix Vector spaces: Vector spaces, linear independence, basis and rank, linear combination of vectors, linearly independent and dependent vectors, rank of a matrix Systems of linear equations:- Method of solution:- Gauss Elimination Method ,Eigenvalues and Eigenvectors and their properties. Utilize the concept related to partial derivatives to solve problems of. (3 marks)

Answer: Begin with the standard definition or identity of *elimination method and evaluate the 5 Applying EigenValues and EigenVectors of a matrix Vector spaces: Vector spaces, linear independence, basis and rank, linear combination of vectors, linearly independent and dependent vectors, rank of a matrix Systems of linear equations:- Method of solution:- Gauss Elimination Method ,Eigenvalues and Eigenvectors and their* *properties. Utilize the concept related to partial derivatives to solve problems of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 4 Applying of functions of two variables Outline the Concept of first and second order. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying of functions of two variables Outline the Concept of first and second order*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Applying Partial derivatives Explain the Concept of maxima and minima of. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying Partial derivatives Explain the Concept of maxima and minima of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain functions of two variables and apply them in 4 Applying related problems. (3 marks)

Answer: Begin with the standard definition or identity of *functions of two variables and apply them in 4 Applying related problems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Demonstrate Multivariate differentiation 3 Applying Calculus: Introduction to Functions, Limits and Continuity, Differentiability, Derivative rules and operations, partial derivatives, maxima and minima of functions of two variables, Multivariate differentiation (Simple problems based on these topics).. (3 marks)

Answer: Begin with the standard definition or identity of *Demonstrate Multivariate differentiation 3 Applying Calculus: Introduction to Functions, Limits and Continuity, Differentiability, Derivative rules and operations, partial derivatives, maxima and minima of functions of two variables, Multivariate differentiation (Simple problems based on these topics).*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 2 Applying space, dependent and independent events Illustrate conditional probability and Bayes’. (3 marks)

Answer: Begin with the standard definition or identity of *2 Applying space, dependent and independent events Illustrate conditional probability and Bayes’*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Applying Theorem Compare the difference between Discrete and. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying Theorem Compare the difference between Discrete and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 3 Applying continuous random variables Calculate the expected value and variance of. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying continuous random variables Calculate the expected value and variance of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 3 Applying discrete and continuous random variables Identify the characteristics of discrete. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying discrete and continuous random variables Identify the characteristics of discrete*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain 1 Understanding performed on data Calculate the measures

of central tendency and. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding performed on data Calculate the measures of central tendency and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Applying dispersion Explain the concept of Sampling and Sampling. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying dispersion Explain the concept of Sampling and Sampling*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding Distributions. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Distributions*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe the types of errors. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the types of errors*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain the concept of vector spaces 2 Understanding Describe linearly independent and dependent**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **4 Applying of functions of two variables Outline the Concept of first and second order**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Applying space, dependent and independent events Illustrate conditional probability and Bayes’**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **1 Understanding performed on data Calculate the measures of central tendency and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No. Website Link
- https://www.w3schools.com/ai/ai_mathematics.asp https://mml-book.github.io/book/mml-book.pdf

IN-PAGE SEARCH

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